

Brutal Series - Paper B16 (2x): Nutrition in Plants / Simple Equations / Forests / Comparing Quantities

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A green leaf makes 0.5 g of glucose each hour of sunlight. On a 9-hour sunny day, half the glucose made is used at once by the plant and the rest is stored as starch. How much starch is stored that day?

[Nutrition in Plants]

- (A) 2.5 g
- (B) 0.45 g
- (C) 2.25 g
- (D) 9 g
- (E) 4.5 g
- (F) 0.25 g
- (G) 1.125 g

2. A young plant's height h (in cm) satisfies $3h - 7 = 2h + 9$. Find h . [Simple Equations]

- (A) 3.2 cm
- (B) 16 cm
- (C) -16 cm
- (D) 2 cm
- (E) 1 cm
- (F) 23 cm
- (G) 8 cm

3. The topmost layer of a forest, formed by the spreading branches of the tallest trees, is called the... [Forests]

- (A) forest floor
- (B) herb layer
- (C) shrub layer
- (D) humus layer
- (E) understorey
- (F) canopy
- (G) root zone

4. A shopkeeper buys a clay plant-pot for Rs 250 and sells it for Rs 300. Find the profit percent. [Comparing

Quantities]

- (A) 16.67%
- (B) 20%
- (C) 80%
- (D) 5%
- (E) 25%
- (F) 50%

5. Of 200 stomata on the lower side of a leaf, 35% are open at noon. At night only 4 stay open. How many more stomata are open at noon than at night? [\[Nutrition in Plants\]](#)
- (A) 74
 - (B) 66
 - (C) 31
 - (D) 130
 - (E) 64
 - (F) 35
 - (G) 70
6. Five times a number, decreased by 8, equals three times the number increased by 10. Find the number. [\[Simple Equations\]](#)
- (A) 9
 - (B) -9
 - (C) 17
 - (D) 1
 - (E) 18
 - (F) 11
 - (G) 2
7. A growing forest absorbs about 22 kg of carbon dioxide per tree each year. A patch has 150 trees. How much carbon dioxide does the patch absorb in one year? [\[Forests\]](#)
- (A) 2200 kg
 - (B) 172 kg
 - (C) 6.8 kg
 - (D) 3300 kg
 - (E) 330 kg
 - (F) 33000 kg
 - (G) 1500 kg
8. Express the ratio 3 : 5 as a percentage of the whole for the first part. [\[Comparing Quantities\]](#)
- (A) 40%
 - (B) 37.5%
 - (C) 53%
 - (D) 35%
 - (E) 30%
 - (F) 60%
 - (G) 62.5%
9. A pitcher plant is green and grows in nitrogen-poor swampy soil, yet it traps and digests insects. Why does it eat insects? [\[Nutrition in Plants\]](#)
- (A) Insects give it the chlorophyll it lacks
 - (B) It is a parasite that lives on the insects
 - (C) It needs the insects' oxygen to breathe
 - (D) It cannot photosynthesise at all so insects are its only food
 - (E) It makes its own glucose by photosynthesis but gets the nitrogen it lacks from insects
 - (F) It traps insects for water, not for food
 - (G) It eats insects only to keep them away from its flowers

- 10.** A nursery plants saplings in equal rows of x . There are 7 full rows plus 5 extra saplings, giving 96 in all. How many saplings are in each row? [\[Simple Equations\]](#)
- (A) 12
 - (B) 101
 - (C) 7
 - (D) 19
 - (E) 14
 - (F) 13
- 11.** Dead leaves and animal droppings on the forest floor are broken down by tiny organisms into a dark, rich substance called... [\[Forests\]](#)
- (A) sap
 - (B) canopy
 - (C) chlorophyll
 - (D) nectar
 - (E) pollen
 - (F) resin
 - (G) humus
- 12.** Find the simple interest on Rs 4000 lent at 8% per annum for 2 years. [\[Comparing Quantities\]](#)
- (A) Rs 4640
 - (B) Rs 640
 - (C) Rs 64
 - (D) Rs 800
 - (E) Rs 320
 - (F) Rs 480
 - (G) Rs 6400
- 13.** Rhizobium bacteria in the root nodules of a bean plant fix nitrogen. If one hectare needs 80 kg of nitrogen and the bacteria supply five-eighths of it, how much nitrogen must the farmer still add? [\[Nutrition in Plants\]](#)
- (A) 75 kg
 - (B) 30 kg
 - (C) 16 kg
 - (D) 45 kg
 - (E) 35 kg
 - (F) 80 kg
- 14.** When 4 is subtracted from one-third of a number, the result is 6. Find the number. [\[Simple Equations\]](#)
- (A) 30
 - (B) 18
 - (C) 20
 - (D) 14
 - (E) 2
 - (F) 10
 - (G) 6

15. In the food chain grass → deer → tiger, the deer is best described as the... [\[Forests\]](#)

- (A) decomposer
- (B) producer
- (C) top carnivore
- (D) saprotroph
- (E) scavenger
- (F) herbivore and primary consumer

16. A price of Rs 800 is increased by 15%. What is the new price? [\[Comparing Quantities\]](#)

- (A) Rs 880
- (B) Rs 680
- (C) Rs 815
- (D) Rs 920
- (E) Rs 1200
- (F) Rs 9200
- (G) Rs 120

17. A leaf has 1.2 g of pigment, of which chlorophyll is 75%. In autumn the chlorophyll breaks down and only the other pigments are left. What mass of pigment remains, and what colour shows? [\[Nutrition in Plants\]](#)

- (A) 0.3 g, yellow-orange
- (B) 1.2 g, yellow-orange
- (C) 0.9 g, green
- (D) 0.45 g, red
- (E) 0.9 g, yellow-orange
- (F) 0.3 g, green
- (G) 0.75 g, yellow-orange

18. Three consecutive whole numbers have a total of 96. Find the greatest of these three numbers. [\[Simple Equations\]](#)

- (A) 30
- (B) 34
- (C) 96
- (D) 35
- (E) 31
- (F) 33
- (G) 32

19. A forest once covered 800 hectares, but 15% was cleared for farming. What area of forest remains? [\[Forests\]](#)

- (A) 720 hectares
- (B) 785 hectares
- (C) 640 hectares
- (D) 920 hectares
- (E) 120 hectares
- (F) 680 hectares

20. In a class of 40 students, 35% are doing a project on plants. How many students are NOT doing the plant project? [\[Comparing Quantities\]](#)

- (A) 14
- (B) 16
- (C) 35
- (D) 26
- (E) 26 percent
- (F) 65

21. Bread kept in a warm, moist place grows a fungus. The fungus has no chlorophyll. How does it get its food? [\[Nutrition in Plants\]](#)

- (A) It makes its own food by photosynthesis
- (B) It steals food directly from a living host
- (C) It absorbs sunlight through the bread
- (D) As a saprotroph: it releases juices onto the bread, breaks it down, then absorbs the dissolved food
- (E) It fixes nitrogen from the air
- (F) It eats the bread whole like an animal

22. A girl is 4 years older than twice her brother's age. Together their ages add up to 28 years. How old is the brother now? [\[Simple Equations\]](#)

- (A) 20 years
- (B) 10 years
- (C) 6 years
- (D) 16 years
- (E) 12 years
- (F) 8 years
- (G) 4 years

23. A forest is often called the 'green lung' of the Earth because it... [\[Forests\]](#)

- (A) filters dust but makes no oxygen
- (B) takes in carbon dioxide and gives out oxygen, keeping the air fresh
- (C) produces all the world's rainfall
- (D) breathes in oxygen and breathes out carbon dioxide
- (E) is shaped like a pair of lungs
- (F) stores rainwater like a sponge only
- (G) is the only place green plants grow

24. A shirt marked at Rs 600 is sold at a 25% discount. Find the selling price. [\[Comparing Quantities\]](#)

- (A) Rs 575
- (B) Rs 750
- (C) Rs 480
- (D) Rs 450
- (E) Rs 150
- (F) Rs 525
- (G) Rs 400

25. To make 1 unit of glucose, a leaf uses 6 units of carbon dioxide and releases 6 units of oxygen. If a plant makes 15 units of glucose in a day, how many units of oxygen does it release? [\[Nutrition in Plants\]](#)

- (A) 15 units
- (B) 30 units
- (C) 6 units
- (D) 21 units
- (E) 90 units
- (F) 45 units
- (G) 36 units

26. A water tank is draining so that its depth after t hours is $(50 - 3t)$ cm. After how many hours is the depth 26 cm? [\[Simple Equations\]](#)

- (A) 16 hours
- (B) 2 hours
- (C) 6 hours
- (D) 25 hours
- (E) 12 hours
- (F) 8 hours

27. Trees release water vapour from their leaves, adding moisture to the air and helping clouds and rain to form. This release of water vapour is called... [\[Forests\]](#)

- (A) transpiration
- (B) respiration
- (C) fertilisation
- (D) germination
- (E) pollination
- (F) photosynthesis

28. Two numbers are in the ratio 4 : 7. If the smaller number is 36, find the larger number. [\[Comparing Quantities\]](#)

- (A) 28
- (B) 49
- (C) 20.57
- (D) 63
- (E) 56
- (F) 45

29. *Cuscuta* (amarbel) is a pale yellow plant with no chlorophyll, seen twined tightly over a hedge. How does it feed? [\[Nutrition in Plants\]](#)

- (A) It is a saprotroph feeding on the dead hedge
- (B) It is a parasite and draws ready-made food from the host plant through suckers
- (C) It makes its own food by photosynthesis
- (D) It absorbs only water, not food, from the hedge
- (E) It lives in symbiosis, helping the hedge in return
- (F) It traps and digests insects on the hedge
- (G) It fixes nitrogen from the air around it

30. One-fifth of a number is equal to 12 less than the number itself. Find the number. [\[Simple Equations\]](#)

- (A) 20
- (B) 48
- (C) 12
- (D) 10
- (E) 3
- (F) 15
- (G) 60

31. A square forest plot has a side of 1200 m. A guard walks all the way around its boundary each morning. How many kilometres does he walk in 5 mornings? [\[Forests\]](#)

- (A) 48 km
- (B) 24 km
- (C) 9.6 km
- (D) 12 km
- (E) 6 km
- (F) 4.8 km

32. A toy bought for Rs 500 is sold for Rs 425. Find the loss percent. [\[Comparing Quantities\]](#)

- (A) 7.5%
- (B) 25%
- (C) 18%
- (D) 15%
- (E) 17.6%
- (F) 12.5%
- (G) 75%

33. During the day a leaf takes in 120 mg of carbon dioxide per hour for photosynthesis, while its cells release 30 mg per hour by respiration. What is the net carbon dioxide taken in per hour, and over 8 daylight hours? [\[Nutrition in Plants\]](#)

- (A) 90 mg/h, 720 mg in 8 h
- (B) 60 mg/h, 480 mg
- (C) 90 mg/h, 540 mg
- (D) 90 mg/h, 720 mg only
- (E) 150 mg/h, 1200 mg
- (F) 30 mg/h, 240 mg

34. A rectangle's length is 3 cm more than its width w . Its perimeter is 38 cm. Find the width. [\[Simple Equations\]](#)

- (A) 8 cm
- (B) 9.5 cm
- (C) 19 cm
- (D) 4 cm
- (E) 16 cm
- (F) 11 cm
- (G) 6 cm

35. One large tree releases about 100 kg of oxygen per year. A village wants a new plantation to give 5000 kg of oxygen a year. How many such trees must they plant? [\[Forests\]](#)

- (A) 20 trees
- (B) 5 trees
- (C) 100 trees
- (D) 500 trees
- (E) 250 trees
- (F) 500000 trees
- (G) 50 trees

36. Rs 5000 is lent at 10% per annum simple interest. What total amount must be paid back after 3 years?

[\[Comparing Quantities\]](#)

- (A) Rs 5150
- (B) Rs 15000
- (C) Rs 6500
- (D) Rs 1500
- (E) Rs 6000
- (F) Rs 5500
- (G) Rs 8000

37. A leaf that is part green and part white (a variegated leaf) is destarched and then left in sunlight, then tested with iodine. Which part turns blue-black, and why? [\[Nutrition in Plants\]](#)

- (A) Neither part, because iodine needs darkness
- (B) Only the green part, because only it has chlorophyll to make starch
- (C) The whole leaf, because starch spreads evenly
- (D) Only the white part, because chlorophyll blocks iodine
- (E) The whole leaf, because light reaches all of it
- (F) Only the white part, because it gets more light
- (G) Only the edges, because they are thinnest

38. If 7 is added to twice a number, the result is 5 less than three times the number. Find the number. [\[Simple](#)

[Equations\]](#)

- (A) 12
- (B) 4
- (C) 6
- (D) 24
- (E) 2
- (F) 1
- (G) -12

39. In a wildlife census a forest's herbivores and carnivores are in the ratio 9:2, and there are 220 of these animals in all. How many carnivores are there? [\[Forests\]](#)

- (A) 180
- (B) 110
- (C) 90
- (D) 40
- (E) 22
- (F) 20
- (G) 44

40. If 25% of a number is 30, what is the number? [\[Comparing Quantities\]](#)

- (A) 55
- (B) 750
- (C) 120
- (D) 60
- (E) 240
- (F) 105
- (G) 7.5

41. A leaf is 60 mm long and 40 mm wide (treat it as a rectangle). 90% of its area is green tissue. What is the green area in square centimetres? [\[Nutrition in Plants\]](#)

- (A) 24 cm²
- (B) 19.2 cm²
- (C) 21.6 cm²
- (D) 216 cm²
- (E) 2160 cm²
- (F) 2.16 cm²
- (G) 240 cm²

42. A pencil costs Rs x and a pen costs Rs $(x + 6)$. Two pencils and three pens together cost Rs 53. Find the cost of one pencil. [\[Simple Equations\]](#)

- (A) Rs 7
- (B) Rs 18
- (C) Rs 10
- (D) Rs 11
- (E) Rs 13
- (F) Rs 5
- (G) Rs 35

43. Why does very little sunlight reach the floor of a thick forest? [\[Forests\]](#)

- (A) Sunlight cannot pass through any air
- (B) The shrubs glow and replace sunlight
- (C) The dense canopy of tall trees blocks most of the light
- (D) The forest floor reflects all the light back up
- (E) Forest trees grow only at night
- (F) The humus layer absorbs the sunlight first
- (G) Roots pull the sunlight underground

44. In a tree survey, 40 out of 250 trees are found to be diseased. What percent of the trees are diseased?

[\[Comparing Quantities\]](#)

- (A) 25%
- (B) 40%
- (C) 10%
- (D) 6.25%
- (E) 160%
- (F) 16 trees
- (G) 16%

45. Before testing a leaf for starch, why is the whole plant kept in the dark for two or three days first? [\[Nutrition](#)

[in Plants\]](#)

- (A) To kill the chlorophyll before the test
- (B) To make the leaf softer for boiling
- (C) To turn the leaf green before the test
- (D) To use up the starch already in the leaves, so any new starch found must be freshly made in the test
- (E) To remove all the water from the leaf
- (F) To fill the leaf with extra starch
- (G) To stop the leaf from breathing

46. Half of a number, increased by 9, equals 20. Find the number. [\[Simple Equations\]](#)

- (A) 11
- (B) 20
- (C) 58
- (D) 31
- (E) 22
- (F) 5.5
- (G) 44

47. A forest fire spreads at a steady 3 km per hour. How far will it spread in 2 hours and 20 minutes? [\[Forests\]](#)

- (A) 4.2 km
- (B) 8 km
- (C) 7 km
- (D) 6.6 km
- (E) 7.6 km
- (F) 6 km
- (G) 9 km

48. A bag of seeds weighs 2.5 kg, and 20% of the seeds are damaged. What mass of seeds is good?

[\[Comparing Quantities\]](#)

- (A) 2.45 kg
- (B) 2.3 kg
- (C) 2 kg
- (D) 2.0 percent
- (E) 0.5 kg
- (F) 1.5 kg
- (G) 0.05 kg

49. After many crops, soil loses nitrogen. A plant has 48 leaves; the ratio of healthy green leaves to yellowing (nitrogen-starved) leaves is 5:1. How many leaves are yellowing? [\[Nutrition in Plants\]](#)

- (A) 5
- (B) 9
- (C) 10
- (D) 6
- (E) 8
- (F) 40
- (G) 48

- 50.** The three angles of a triangle are x , $2x$ and $3x$ degrees. Using the fact that they add up to 180 degrees, find x . [\[Simple Equations\]](#)
- (A) 90 degrees
 - (B) 18 degrees
 - (C) 45 degrees
 - (D) 36 degrees
 - (E) 60 degrees
 - (F) 30 degrees
- 51.** Forests help prevent floods mainly because their roots and humus... [\[Forests\]](#)
- (A) make the rain fall somewhere else
 - (B) turn rainwater into oxygen
 - (C) heat the rainwater until it evaporates
 - (D) freeze the rainwater in the ground
 - (E) stop rain from falling on the forest
 - (F) push the water straight into rivers faster
 - (G) hold the soil together and soak up rainwater, slowing the run-off
- 52.** Rs 1200 is shared between A and B in the ratio 5 : 3. How much does B receive? [\[Comparing Quantities\]](#)
- (A) Rs 450
 - (B) Rs 400
 - (C) Rs 480
 - (D) Rs 360
 - (E) Rs 150
 - (F) Rs 750
 - (G) Rs 600
- 53.** A leaf stores 0.4 g of starch on a sunny day but only 0.1 g on a cloudy day. In one week there are 4 sunny days and 3 cloudy days. How much starch is stored that week? [\[Nutrition in Plants\]](#)
- (A) 2.1 g
 - (B) 1.9 g
 - (C) 2.8 g
 - (D) 1.3 g
 - (E) 0.3 g
 - (F) 0.5 g
 - (G) 1.6 g
- 54.** When a number is divided by 4 and then 3 is added, the result is 8. Find the number. [\[Simple Equations\]](#)
- (A) 44
 - (B) 17
 - (C) 2
 - (D) 5
 - (E) 20
 - (F) 32

55. Arrange the layers of a forest from the ground upward. [\[Forests\]](#)
- (A) Herbs and shrubs, then understorey trees, then the canopy at the top
 - (B) Humus, then canopy, then understorey
 - (C) Shrubs, then humus, then canopy
 - (D) Canopy, then understorey, then herbs at the top
 - (E) Canopy at the bottom and herbs at the top
 - (F) Understorey, then canopy, then herbs
 - (G) Herbs, then canopy, then understorey
56. The population of a town rises from 8000 to 8600 in a year. Find the percent increase. [\[Comparing Quantities\]](#)
- (A) 7%
 - (B) 6.98%
 - (C) 8.6%
 - (D) 600%
 - (E) 75%
 - (F) 6%
 - (G) 7.5%
57. Air is about 21% oxygen. A sealed jar holds 500 mL of air. During the day a green plant inside adds oxygen equal to 4% of the jar's volume. What is the new volume of oxygen in the jar? [\[Nutrition in Plants\]](#)
- (A) 121 mL
 - (B) 20 mL
 - (C) 145 mL
 - (D) 25 mL
 - (E) 105 mL
 - (F) 125 mL
 - (G) 100 mL
58. In a forest, deer and the tigers that hunt them number 90 animals together. There are 14 times as many deer as tigers. How many tigers are there? [\[Simple Equations\]](#)
- (A) 15
 - (B) 84
 - (C) 7
 - (D) 9
 - (E) 6
 - (F) 5
59. A nursery plants 1 sapling in every 0.5 square metre of ground. How many saplings fit in a plot 20 m long and 15 m wide? [\[Forests\]](#)
- (A) 60 saplings
 - (B) 450 saplings
 - (C) 300 saplings
 - (D) 600 saplings
 - (E) 150 saplings
 - (F) 35 saplings
 - (G) 1200 saplings

60. A trader wants to gain 12% on an item that costs Rs 250. At what price must it be sold? [\[Comparing Quantities\]](#)

- (A) Rs 262
- (B) Rs 312
- (C) Rs 280
- (D) Rs 220
- (E) Rs 300
- (F) Rs 30

61. Lichens are an alga and a fungus living together: the alga makes food and the fungus supplies water and shelter, and both benefit. This partnership is called... [\[Nutrition in Plants\]](#)

- (A) decomposition
- (B) photosynthesis
- (C) symbiosis
- (D) insectivory
- (E) host-feeding
- (F) parasitism
- (G) saprotrophic nutrition

62. Subtracting 5 from four times a number gives 31. Find the number. [\[Simple Equations\]](#)

- (A) 4
- (B) 36
- (C) 18
- (D) 9.25
- (E) 26
- (F) 9
- (G) 6.5

63. Vultures, ants and other animals that feed on dead bodies and waste, helping to keep the forest clean, are called... [\[Forests\]](#)

- (A) scavengers
- (B) autotrophs
- (C) pollinators
- (D) producers
- (E) top predators
- (F) herbivores
- (G) parasites

64. What percentage is the fraction $\frac{7}{20}$ equal to? [\[Comparing Quantities\]](#)

- (A) 3.5%
- (B) 20%
- (C) 70%
- (D) 7%
- (E) 13%
- (F) 35%
- (G) 27%

65. A destarched leaf is half-covered with foil and left in sunlight for 6 hours. The exposed half is 30 cm² and makes starch at 0.2 g per 10 cm² each hour. How much starch does the exposed half make in all?

[Nutrition in Plants]

- (A) 1.2 g
- (B) 0.6 g
- (C) 6 g
- (D) 2.4 g
- (E) 3.6 g
- (F) 0.36 g

66. Three less than half of a number equals one-fourth of the number plus 2. Find the number. [Simple

Equations]

- (A) 20
- (B) 12
- (C) 5
- (D) 40
- (E) 4
- (F) 10

67. A forest department aims to plant 2500 trees and has finished 60% of the work. How many trees are still to be planted? [Forests]

- (A) 60 trees
- (B) 2440 trees
- (C) 900 trees
- (D) 1500 trees
- (E) 1000 trees
- (F) 500 trees

68. Rs 2000 earns Rs 240 as simple interest in 2 years. Find the rate of interest per annum. [Comparing

Quantities]

- (A) 4.8%
- (B) 6%
- (C) 8%
- (D) 240%
- (E) 5%
- (F) 12%
- (G) 3%

69. Which set lists ONLY the raw materials a green leaf takes IN for photosynthesis (not the products)?

[Nutrition in Plants]

- (A) Water, oxygen and glucose
- (B) Oxygen, carbon dioxide and starch
- (C) Carbon dioxide, water, sunlight and chlorophyll
- (D) Glucose, starch and chlorophyll
- (E) Glucose, oxygen and water
- (F) Carbon dioxide, glucose and oxygen
- (G) Oxygen, water and sunlight

70. The sum of two numbers is 40, and one number is 4 times the other. Find the larger number. [\[Simple](#)

[Equations\]](#)

- (A) 8
- (B) 36
- (C) 5
- (D) 10
- (E) 30
- (F) 32

71. The wide variety of different plants and animals living together in a forest is called its... [\[Forests\]](#)

- (A) transpiration
- (B) biodiversity
- (C) canopy
- (D) food chain
- (E) humus
- (F) population
- (G) ecosystem boundary

72. A 60-litre tank is 35% full. How many more litres are needed to fill it completely? [\[Comparing Quantities\]](#)

- (A) 60 litres
- (B) 42 litres
- (C) 35 litres
- (D) 39 litres
- (E) 25 litres
- (F) 21 litres

73. Two correct ways to put nitrogen back into worn-out soil are... [\[Nutrition in Plants\]](#)

- (A) Keeping the field dark for a few days
- (B) Adding more carbon dioxide to the air
- (C) Removing all weeds and burning leaves
- (D) Adding lime and ploughing deeper
- (E) Spraying the leaves with iodine
- (F) Adding manure or fertiliser, and growing leguminous plants that host Rhizobium

74. Water in a beaker is heated so its temperature is $(25 + 5t)$ degrees after t minutes. After how many minutes does it reach 80 degrees? [\[Simple Equations\]](#)

- (A) 13 minutes
- (B) 21 minutes
- (C) 16 minutes
- (D) 9 minutes
- (E) 55 minutes
- (F) 11 minutes

75. Humus makes soil fertile. If 8 kg of dead leaves produce 2 kg of humus, how much humus comes from 28 kg of dead leaves at the same rate? [\[Forests\]](#)

- (A) 22 kg
- (B) 4 kg
- (C) 7 kg
- (D) 9 kg
- (E) 3.5 kg
- (F) 14 kg
- (G) 112 kg

76. A fruit seller buys 12 mangoes for Rs 120 and sells them at Rs 12 each. Find the profit percent.

[\[Comparing Quantities\]](#)

- (A) 144%
- (B) 8%
- (C) 12%
- (D) 24%
- (E) 20%
- (F) 2%
- (G) 10%

77. Mushrooms and yeast cannot make their own food and grow on dead or decaying matter. Based on how they feed, they are best called... [\[Nutrition in Plants\]](#)

- (A) host plants
- (B) autotrophs
- (C) insectivores
- (D) parasites
- (E) symbionts
- (F) producers
- (G) saprotrophs

78. Solve for x: $2(x - 4) = 3(x - 7)$. [\[Simple Equations\]](#)

- (A) 5
- (B) 13
- (C) 3
- (D) 29
- (E) 9
- (F) 11
- (G) -13

79. Why is a forest described as a 'dynamic, living entity'? [\[Forests\]](#)

- (A) Nothing in it ever changes over time
- (B) It exists only in storybooks
- (C) It is made only of non-living soil and rock
- (D) It is always changing and full of life - plants grow, animals interact, and matter is recycled
- (E) It is a single giant living animal
- (F) It is alive only during the rainy season
- (G) It can get up and move to a new place

80. In how many years will Rs 1000 earn Rs 300 as simple interest at a rate of 6% per annum? [\[Comparing Quantities\]](#)

- (A) 5 years
- (B) 2 years
- (C) 10 years
- (D) 50 years
- (E) 6 years
- (F) 4 years
- (G) 3 years

81. Water absorbed by the roots must reach the leaves for photosynthesis. It travels up the plant mainly through... [\[Nutrition in Plants\]](#)

- (A) the chlorophyll in the cells
- (B) the stomata on the leaves
- (C) the air spaces in the soil
- (D) the flowers and fruits
- (E) the phloem carrying glucose down
- (F) tube-like vessels (xylem) in the roots and stem
- (G) the cell walls of the bark

82. A father is three times as old as his son. In 5 years the father will be 50. How old is the son now? [\[Simple Equations\]](#)

- (A) 16 years
- (B) 45 years
- (C) 20 years
- (D) 15 years
- (E) 10 years
- (F) 18 years
- (G) 12 years

83. Of all the rain that falls on a forest, about 70% soaks into the ground. If 900 mm of rain falls in a year, how much soaks in? [\[Forests\]](#)

- (A) 270 mm
- (B) 790 mm
- (C) 630 mm
- (D) 70 mm
- (E) 830 mm
- (F) 450 mm

84. Convert the improper fraction $\frac{5}{4}$ into a percentage. [\[Comparing Quantities\]](#)

- (A) 1.25%
- (B) 80%
- (C) 125%
- (D) 20%
- (E) 125 percent of nothing
- (F) 45%

85. A potato is an underground stem that is swollen with stored food. Why does the plant store food in it?

[Nutrition in Plants]

- (A) The potato fixes nitrogen for the plant
- (B) The food is waste the plant cannot use
- (C) Stored food keeps insects away from the roots
- (D) The potato traps insects to get its food
- (E) The potato absorbs ready-made food from the soil
- (F) Food made in the leaves by photosynthesis is moved there and stored for later use
- (G) The potato makes its own food by photosynthesis underground

86. Adding 12 to a number gives the same result as subtracting that number from 40. Find the number.

[Simple Equations]

- (A) 26
- (B) 14
- (C) 7
- (D) 52
- (E) 28
- (F) 2

87. If all the deer in a forest were to die out, the most likely immediate effect on the tigers would be that they... [Forests]

- (A) lose a major source of food, so their numbers fall
- (B) increase quickly in number
- (C) leave to live in the ocean
- (D) are completely unaffected
- (E) grow much larger in size
- (F) turn into herbivores and eat grass

88. Boys and girls planting trees are in the ratio 7 : 5. If there are 84 children in all, how many are girls?

[Comparing Quantities]

- (A) 30
- (B) 35
- (C) 42
- (D) 60
- (E) 12
- (F) 7

89. Gases enter and leave a leaf through tiny pores called stomata, and each pore is opened or closed by a pair of... [Nutrition in Plants]

- (A) veins
- (B) chlorophyll grains
- (C) xylem vessels
- (D) root hairs
- (E) sieve tubes
- (F) guard cells
- (G) air sacs

90. Five times a number equals that number plus 36. Find the number. [\[Simple Equations\]](#)

- (A) 6
- (B) 18
- (C) 45
- (D) 12
- (E) 36
- (F) 9
- (G) 4

91. A rectangular forest reserve is 6 km long and 4 km wide. A 1 km wide protective buffer runs all along the inside of the boundary. What is the area of the inner core (non-buffer) zone? [\[Forests\]](#)

- (A) 10 km²
- (B) 12 km²
- (C) 20 km²
- (D) 15 km²
- (E) 8 km²
- (F) 6 km²

92. An item marked at Rs 1500 is offered at 20% off. How much money is saved by buying it on offer?

[\[Comparing Quantities\]](#)

- (A) Rs 1480
- (B) Rs 30
- (C) Rs 1200
- (D) Rs 200
- (E) Rs 300
- (F) Rs 750

93. A green plant making food from simple substances is called a producer. In a food chain it is placed first because... [\[Nutrition in Plants\]](#)

- (A) it lives the longest of all
- (B) it eats the animals above it
- (C) it is the only living thing that breathes
- (D) it makes the food that all other living things in the chain depend on
- (E) it is the largest organism in the chain
- (F) it gets food ready-made from the soil

94. The length of a bean plant is given by $L = 5 + 2n$ cm, where n is the number of weeks. In which week will the plant be 23 cm long? [\[Simple Equations\]](#)

- (A) Week 18
- (B) Week 11
- (C) Week 9
- (D) Week 4
- (E) Week 14
- (F) Week 7
- (G) Week 28

95. Birds, bees and other animals help a forest grow chiefly by... [\[Forests\]](#)

- (A) drinking up the forest's rainwater
- (B) stopping all the trees from growing
- (C) spreading seeds and pollen, which helps new plants form
- (D) making humus directly from rock
- (E) turning carbon dioxide into soil
- (F) blocking sunlight from the canopy
- (G) eating the roots of every tree

96. A plant that is 50 cm tall grows 8% taller during one season. How tall is it at the end of the season?

[\[Comparing Quantities\]](#)

- (A) 46 cm
- (B) 4 cm
- (C) 54 cm
- (D) 52 cm
- (E) 50.08 cm
- (F) 58 cm
- (G) 62 cm

97. In photosynthesis a leaf uses 6 units of carbon dioxide and 6 units of water to make glucose. If a plant has 6 units of carbon dioxide but only 4 units of water available, what limits how much glucose it can make?

[\[Nutrition in Plants\]](#)

- (A) Water, because there is not enough of it for the carbon dioxide present
- (B) Neither - it makes the full amount anyway
- (C) Chlorophyll, which is also a raw material used up
- (D) Starch, which must be added in
- (E) Sunlight, which the question removed
- (F) Oxygen, because none is supplied

98. The sum of a number and one-third of itself is 28. Find the number. [\[Simple Equations\]](#)

- (A) 84
- (B) 9.33
- (C) 42
- (D) 7
- (E) 21
- (F) 36

99. A young teak tree adds about 2 cm to the girth (distance around) of its trunk each year. If its girth is now 18 cm, what will the girth be after 6 years? [\[Forests\]](#)

- (A) 108 cm
- (B) 26 cm
- (C) 36 cm
- (D) 30 cm
- (E) 20 cm
- (F) 12 cm
- (G) 42 cm

100. By selling a basket for Rs 360, a vendor makes a profit of 20%. Find the cost price of the basket.

[Comparing Quantities]

- (A) Rs 432
- (B) Rs 300
- (C) Rs 288
- (D) Rs 340
- (E) Rs 72
- (F) Rs 450
- (G) Rs 320